

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

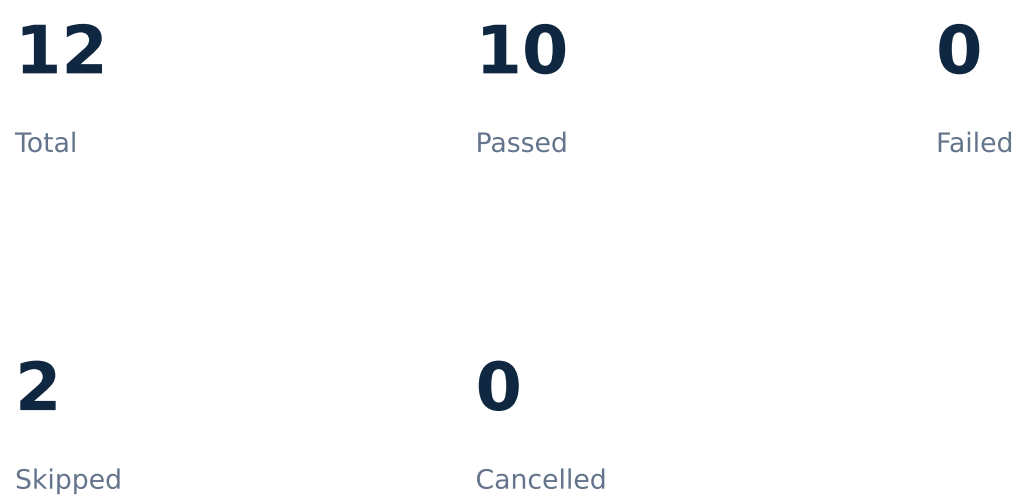
20260813_234502_0efc01

Date

2026-08-13T22:45:02.978478+00:00

Executive Summary

20260813_234502_0efc01



Overall measured availability: 45.58%

During this campaign, 10 of 12 scenarios passed, 0 failed and 2 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes	swarm
container-kill	1	1
cpu	1	1
latency	1	1
memory	1	1
network-partition	1	1
node-failure	1	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
container-kill	swarm	3.125	12.656	True	True	11.586	21.983	11	15.38	PASS	N/A	N/A
node-failure	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Single-node Docker Swarm cannot demonstrate real node	N/A
cpu	swarm	N/A	N/A	N/A	Not Required	13.079	21.954	N/A	100.0	PASS	N/A	N/A
memory	swarm	N/A	N/A	N/A	Not Required	11.919	21.455	N/A	100.0	PASS	N/A	N/A
network-partition	swarm	0.094	0.015	True	True	19.601	57.526	15	N/A	PASS	N/A	N/A
latency	swarm	4.562	5.015	True	True	4521.012	4538.434	11	N/A	PASS	N/A	N/A
container-kill	kubernetes	6.093	19.015	True	True	1361.458	2046.665	7	22.22	PASS	N/A	N/A
node-failure	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL	N/A
cpu	kubernetes	N/A	N/A	N/A	Not Required	26.973	65.227	N/A	100.0	PASS	N/A	N/A
memory	kubernetes	N/A	N/A	N/A	Not Required	16.527	20.741	N/A	100.0	PASS	N/A	N/A
network-partition	kubernetes	0.172	0.109	True	True	31.339	103.623	15	N/A	PASS	N/A	N/A
latency	kubernetes	4.579	5.016	True	True	4527.381	4543.487	9	18.18	PASS	N/A	N/A

swarm — container-kill

State: **Pass**
Detection Time (s): 3.125
Recovery Time (s): 12.656

HTTP Success: N/A

HTTP Failures: 11

Availability (%): 15.38

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — node-failure

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Single-node Docker Swarm cannot demonstrate real node rescheduling.

swarm — cpu

Status: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — memory

State: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — network-partition

State: PASS
Detection Time (s): 0.094

Recovery Time (s): 0.015

HTTP Success: N/A

HTTP Failures: 15

Availability (%): N/A

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — latency

State: Pass
Detection Time (s): 4.562

Recovery Time (s): 5.015

HTTP Success: N/A

HTTP Failures: 11

Availability (%): N/A

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — container-kill

State: Pass
Detection Time (s): 6.093

Recovery Time (s): 19.015

HTTP Success: N/A

HTTP Failures: 7

Availability (%): 22.22

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — node-failure

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

kubernetes — cpu

Status: Pass
Detection Time (s): N/A
Recovery Time (s): N/A
HTTP Success: N/A
HTTP Failures: N/A
Availability (%): 100.0
CPU (%): N/A
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — memory

State: Pass

Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — network-partition

Status: PASS
Detection Time (s): 0.172

Recovery Time (s): 0.109

HTTP Success: N/A

HTTP Failures: 15

Availability (%): N/A

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — latency

Status: Pass
Detection Time (s): 4.579

Recovery Time (s): 5.016

HTTP Success: N/A

HTTP Failures: 9

Availability (%): 18.18

CPU (%): N/A

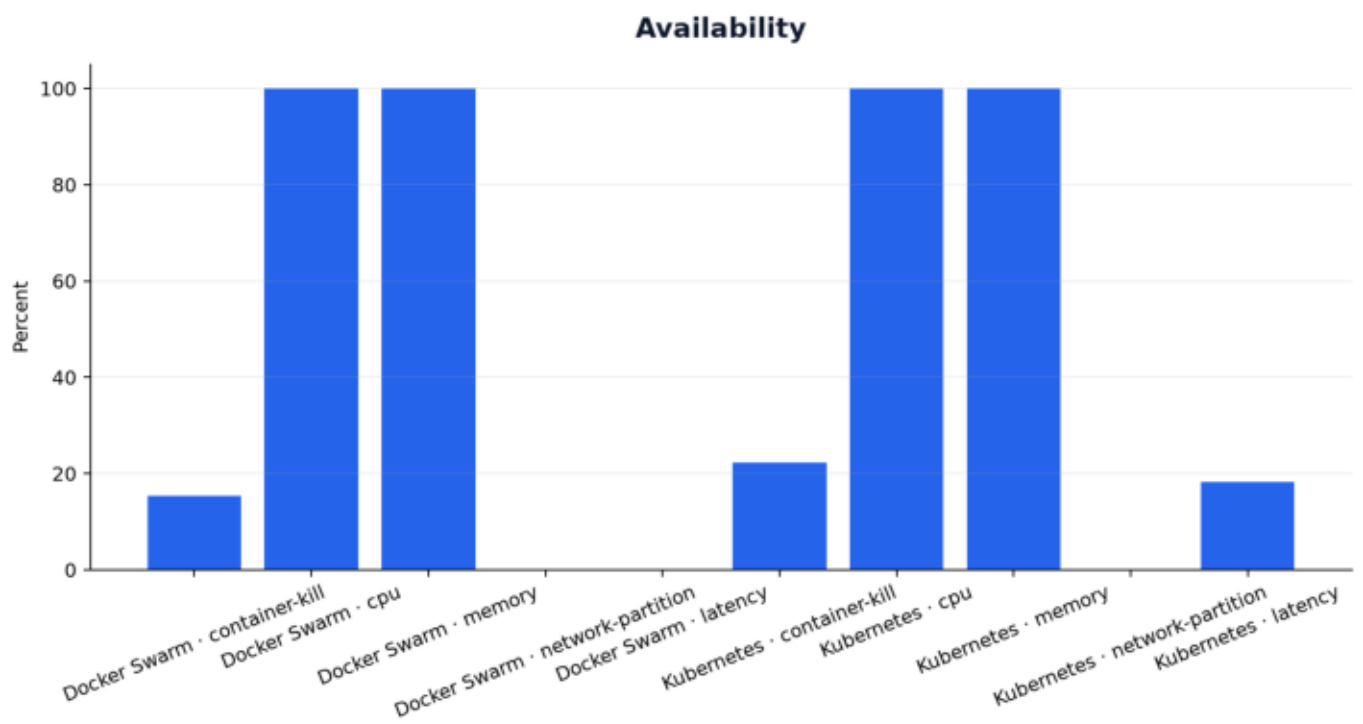
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

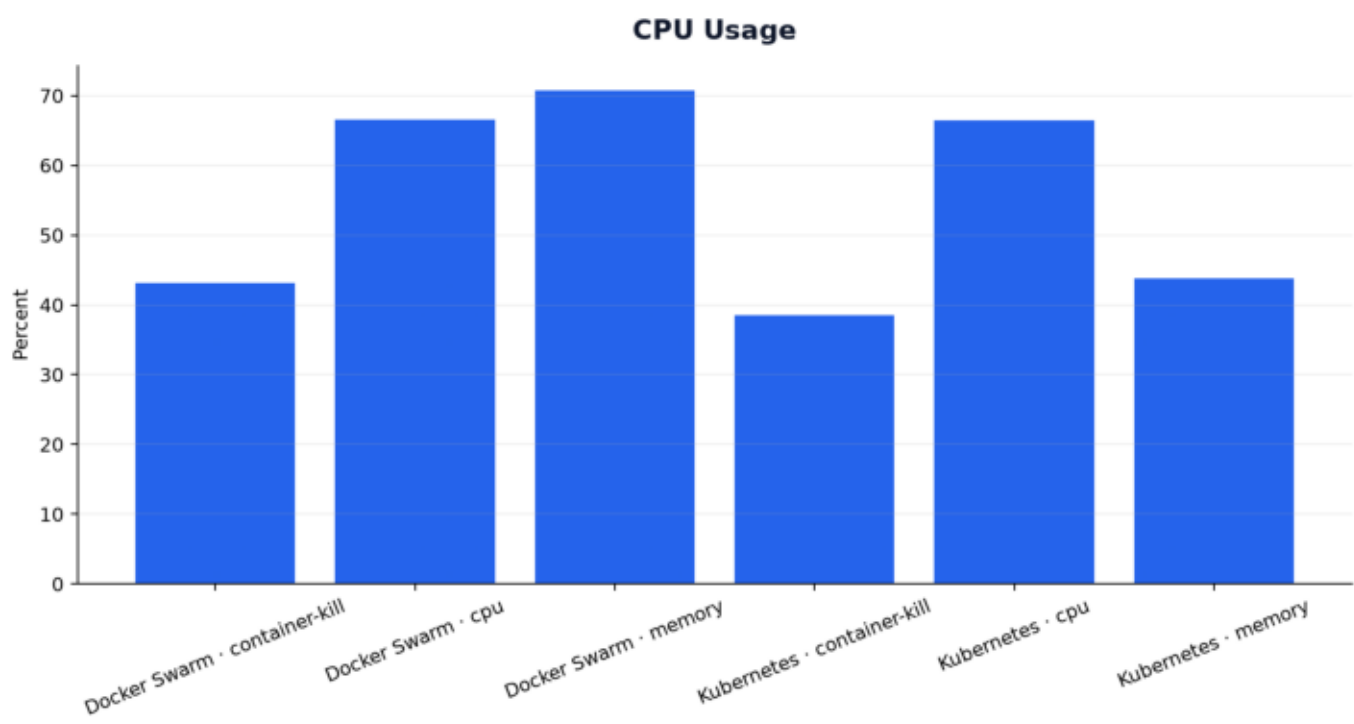
Availability

Current-session measured data



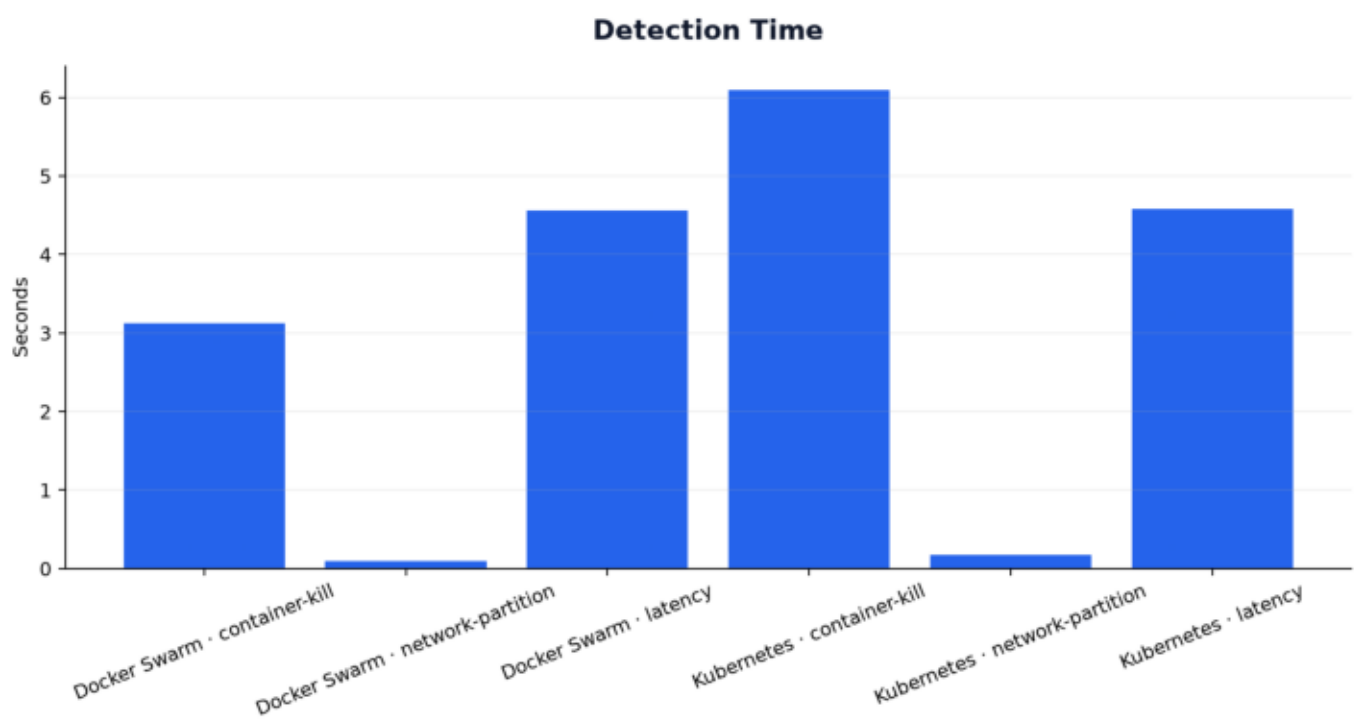
Cpu Usage

Current-session measured data



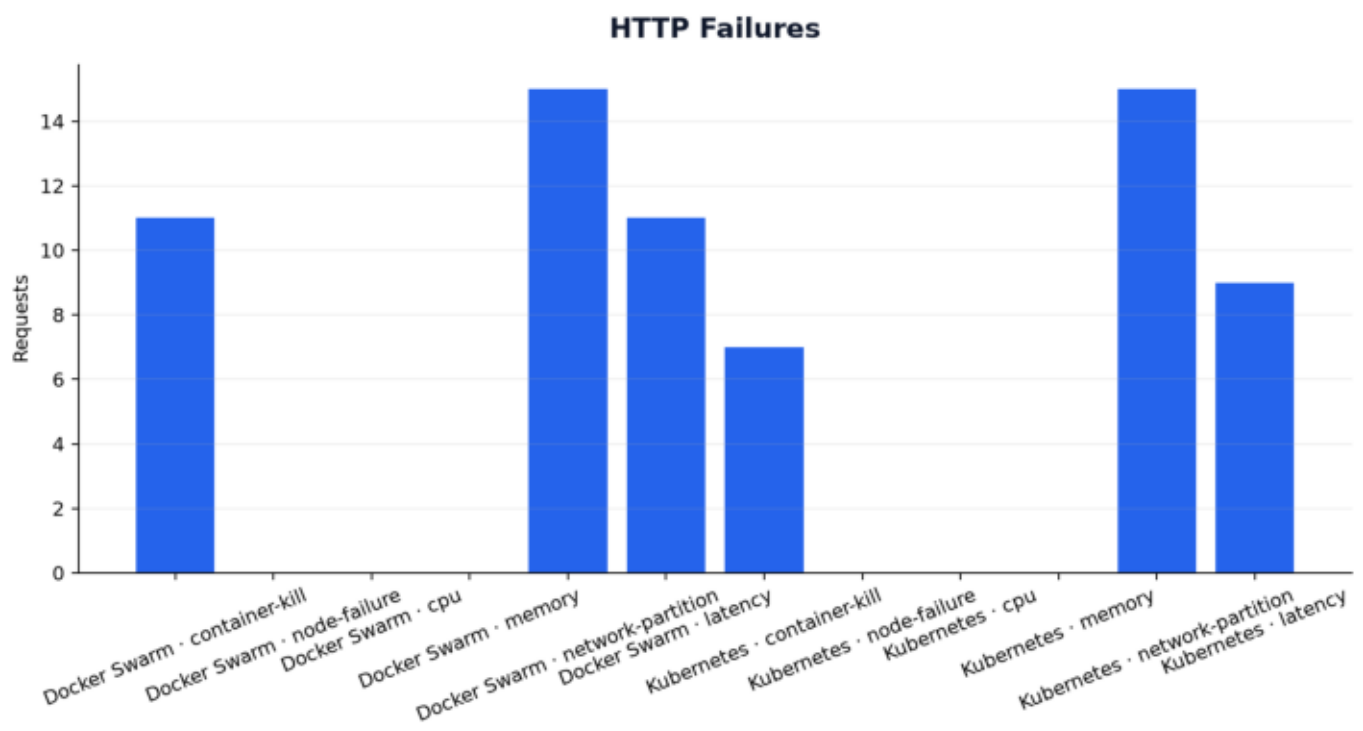
Detection Time

Current-session measured data



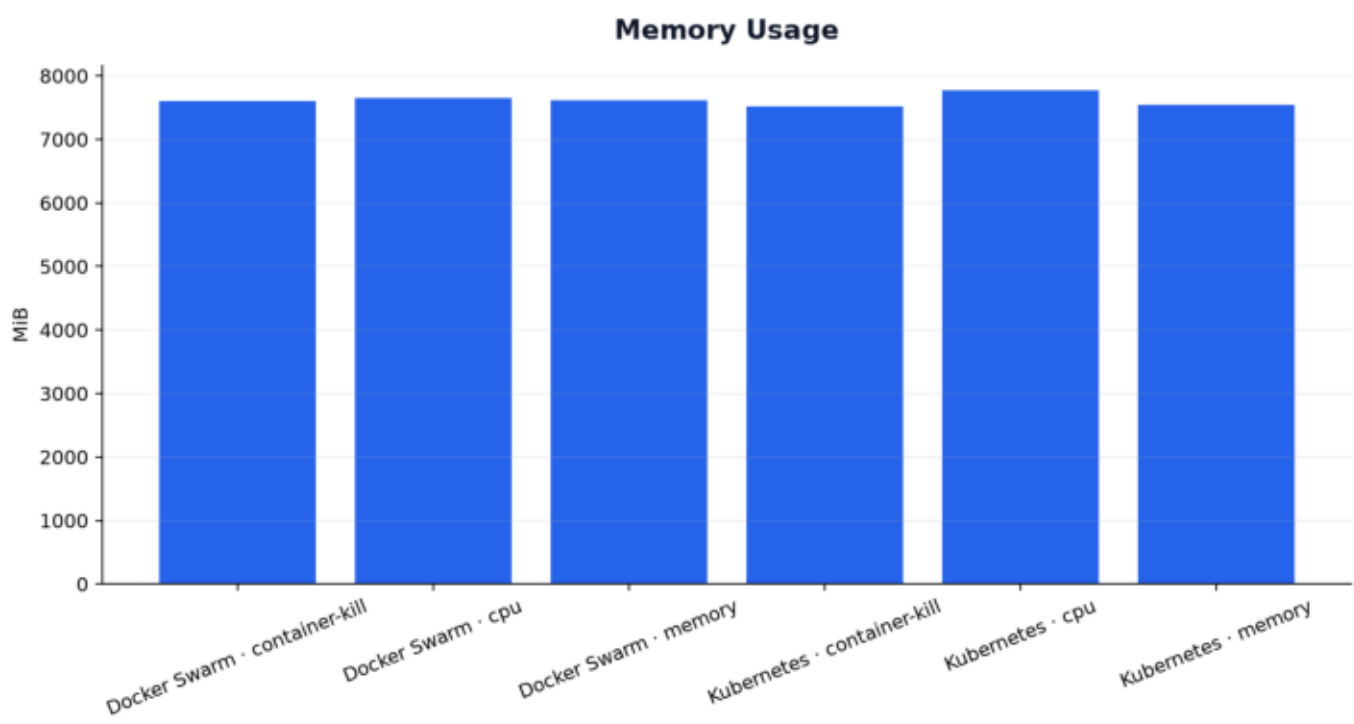
Http Failures

Current-session measured data



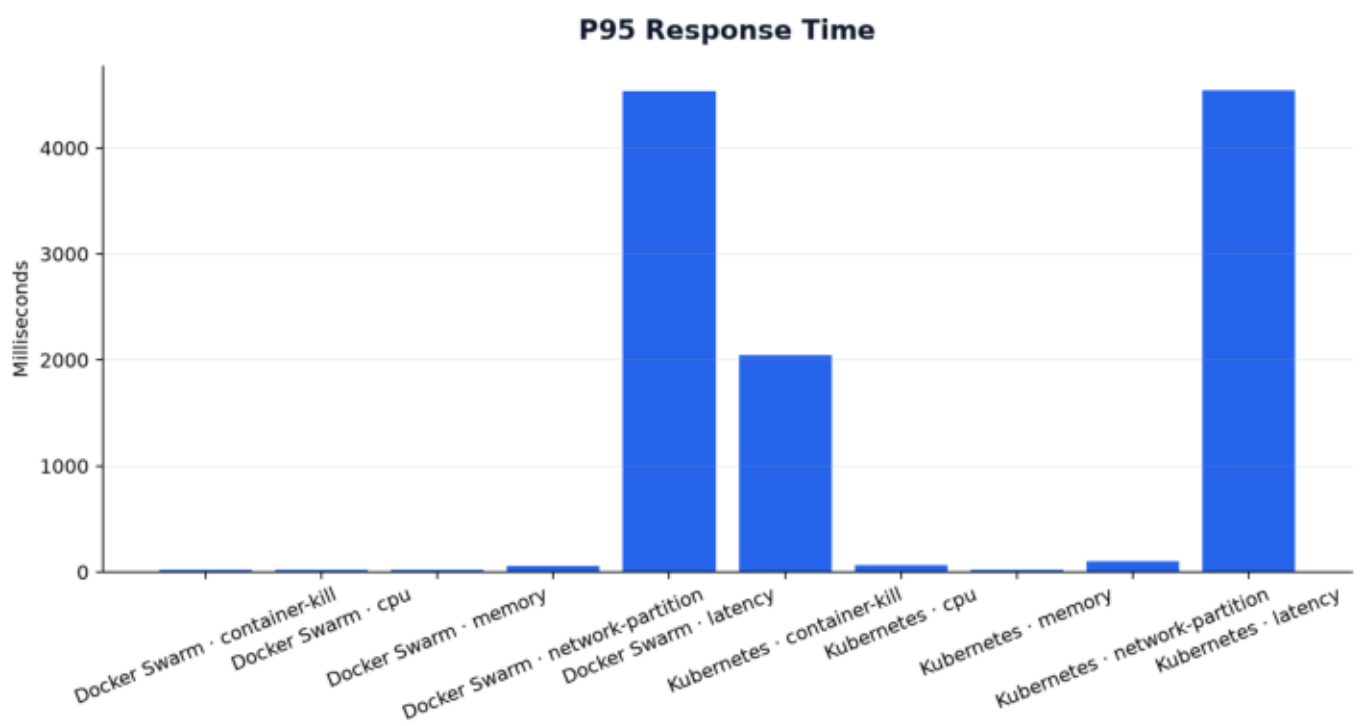
Memory Usage

Current-session measured data



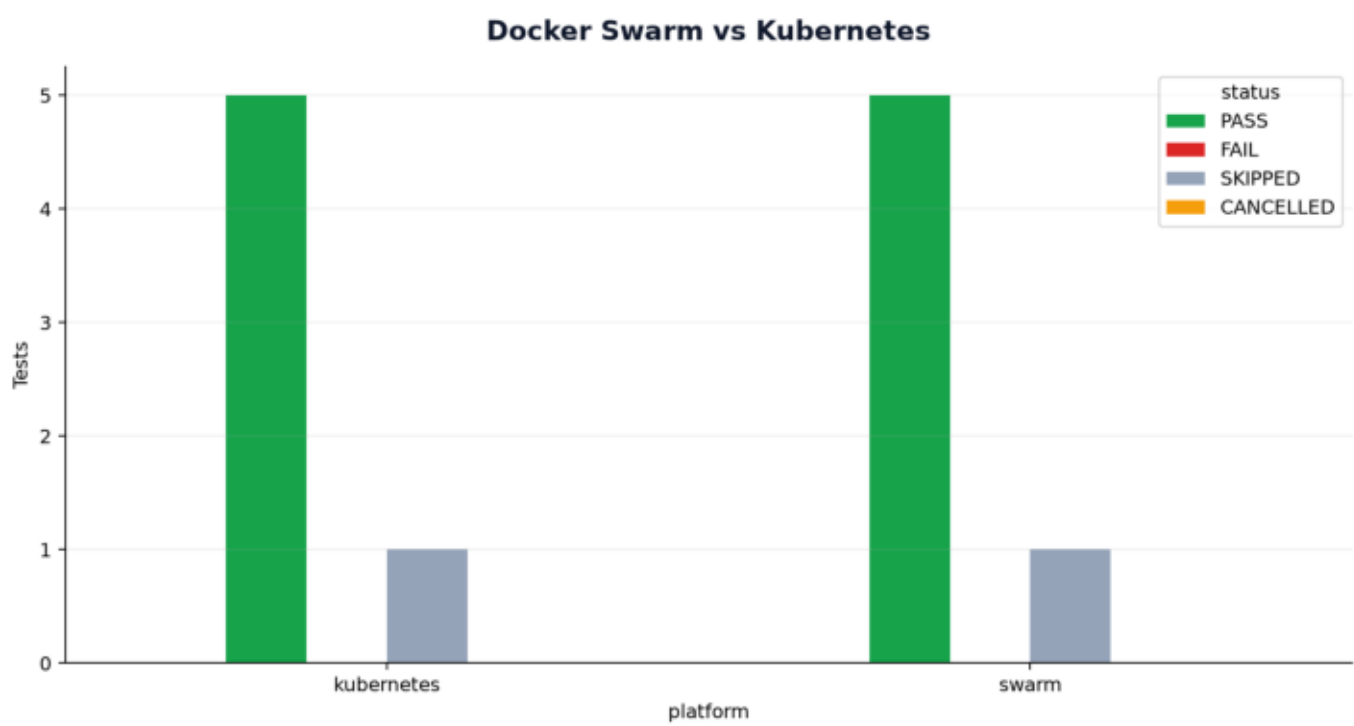
P95 Response Time

Current-session measured data



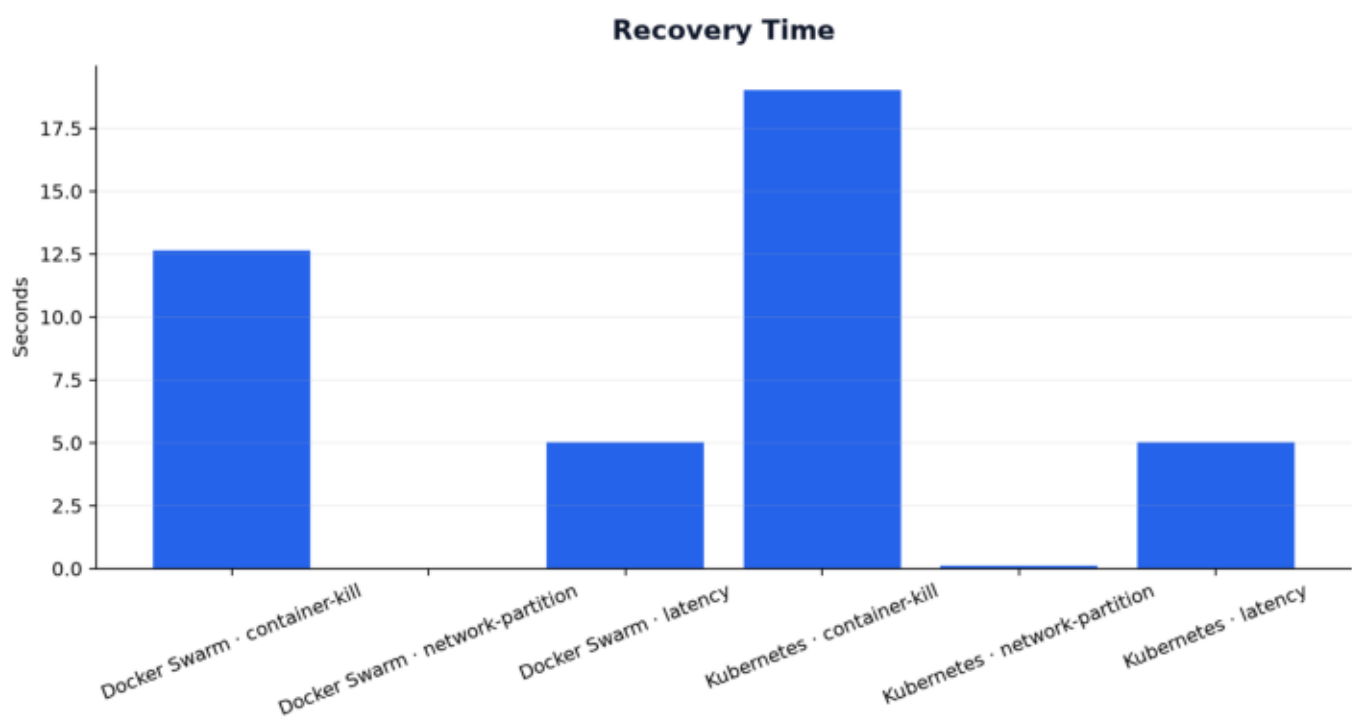
Platform Comparison

Current-session measured data



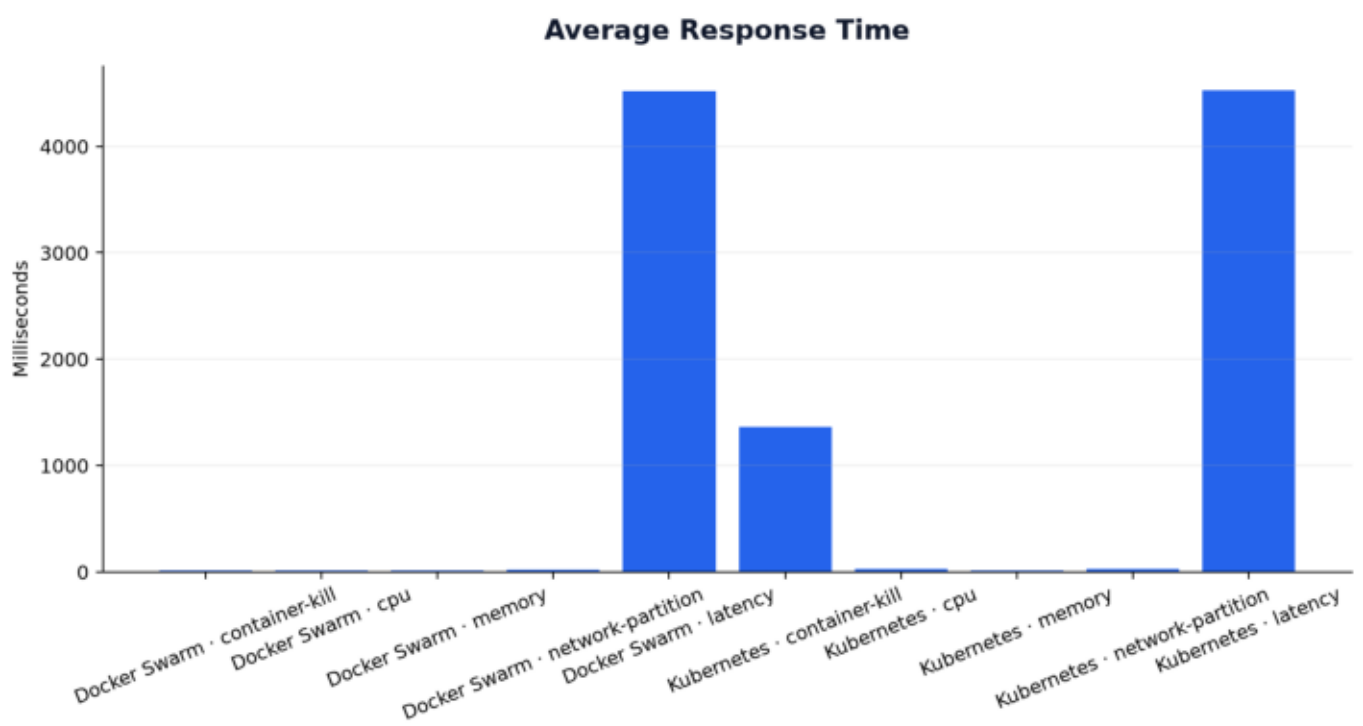
Recovery Time

Current-session measured data



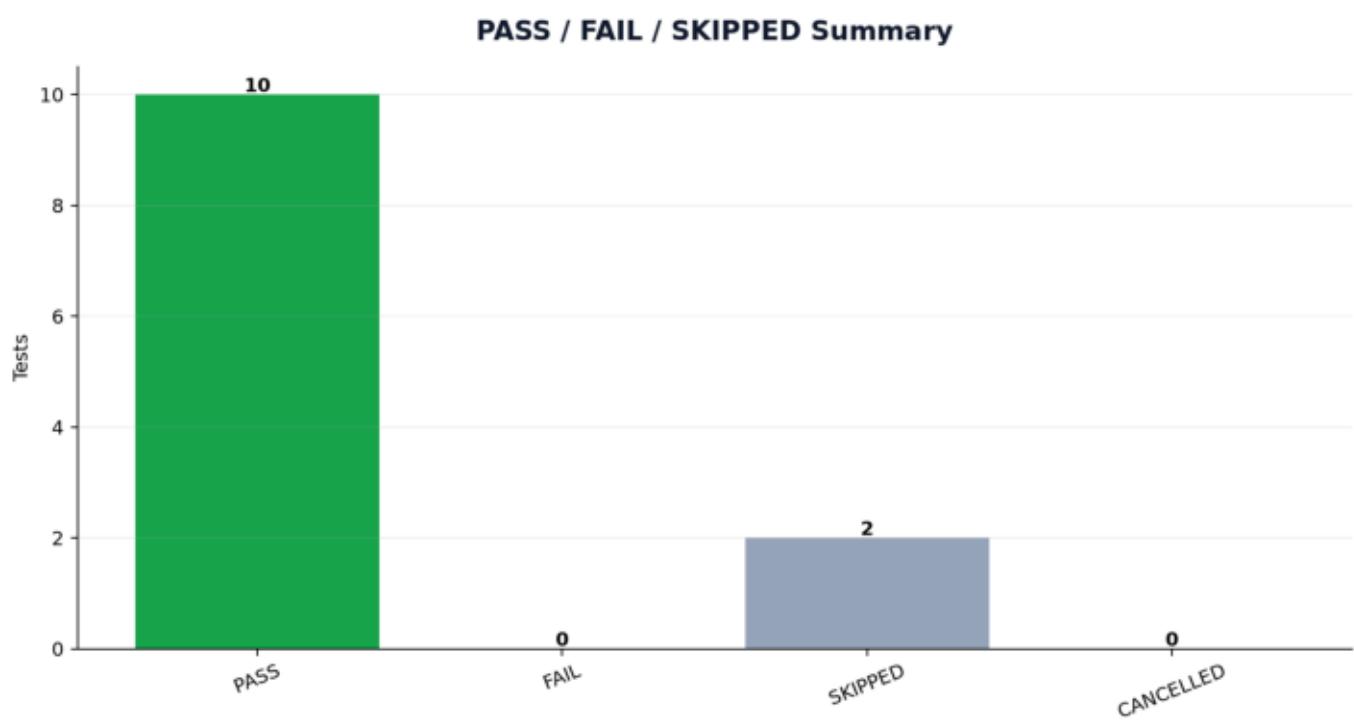
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
—	—	—	No failures recorded.

Skipped Tests

Platform	Scenario	Exact reason
swarm	node-failure	Single-node Docker Swarm cannot demonstrate real node rescheduling.
kubernetes	node-failure	Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

Swarm vs Kubernetes Comparison

Platform	PASS	SKIPPED
kubernetes	5	1
swarm	5	1

Final Conclusion

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During this campaign, 10 of 12 scenarios passed, 0 failed and 2 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.